

English

Computational Thinking

Detective

Dalton & Terru¹

"Missing Cat Tom"

Teacher's
Guide

Create Your Universe!
CREVERS



Goal

Find out cat's characteristics such as eyes, nose, hat, and etc. and figure out how to collect data.

How

Encourage to ask questions about cat's appearance. Explain children to find cat's characteristics such as eyes, nose, hat, and etc.

Example

"Cat's eyes are green. Please find green eyes."

※ Required tools : pen or pencil

Application play

Draw a montage of another cat by combining characteristics of the cat in many ways.

Goal

Learn how to a abstract information by making montage flyers of cat.

How

Draw a montage with cat's characteristics identified on page 10.

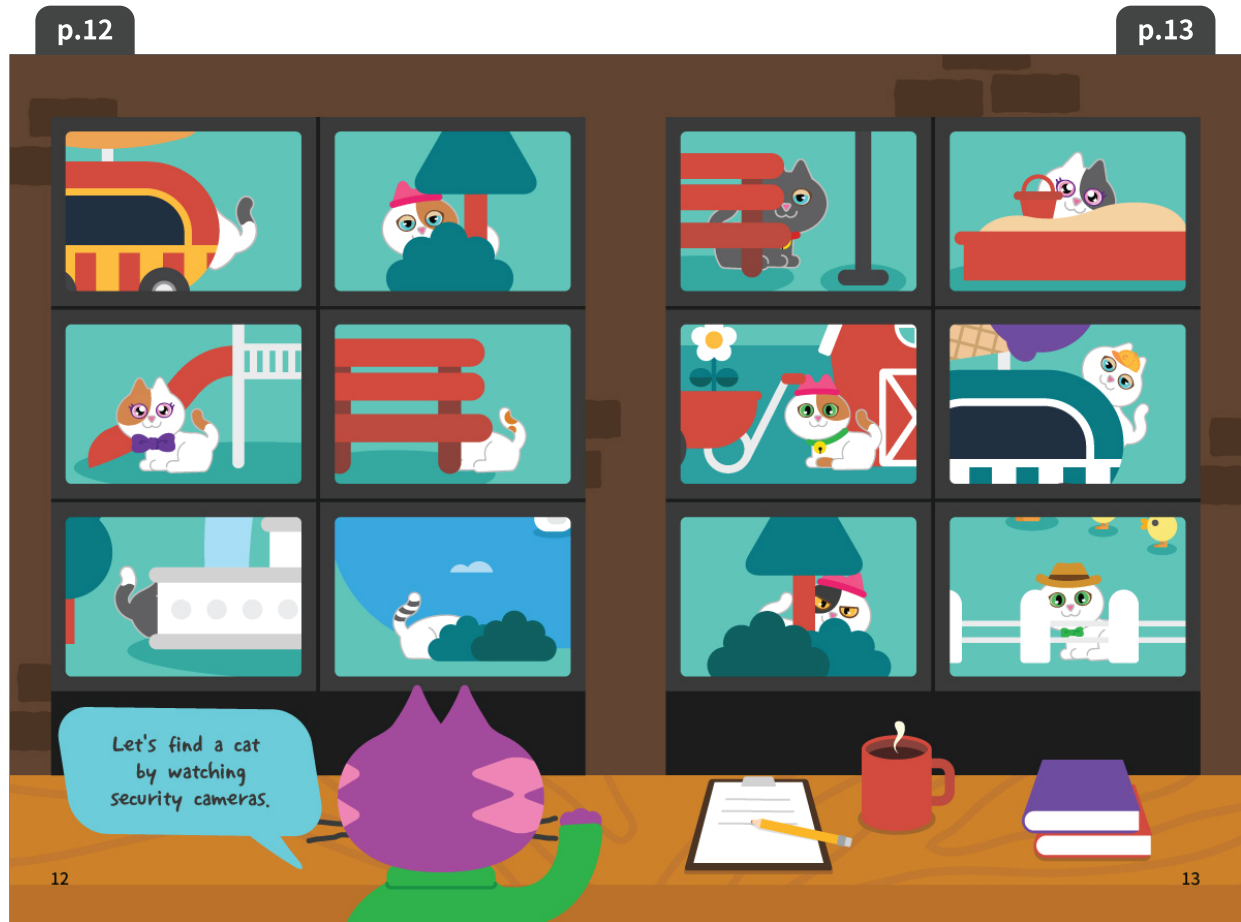
Example

"The cat has pretty green eyes."



Find and draw green pupils of eyes ② on page 10.

※ Required tools : colored pencils or crayons



Goal

Learn how to figure out characteristics of cat and compare cats in the pictures.

How

Remind appearance of answer cat again on p.11, talk about which part is different by comparing with other cats taken partially on security camera, and find the missing cat.

Example

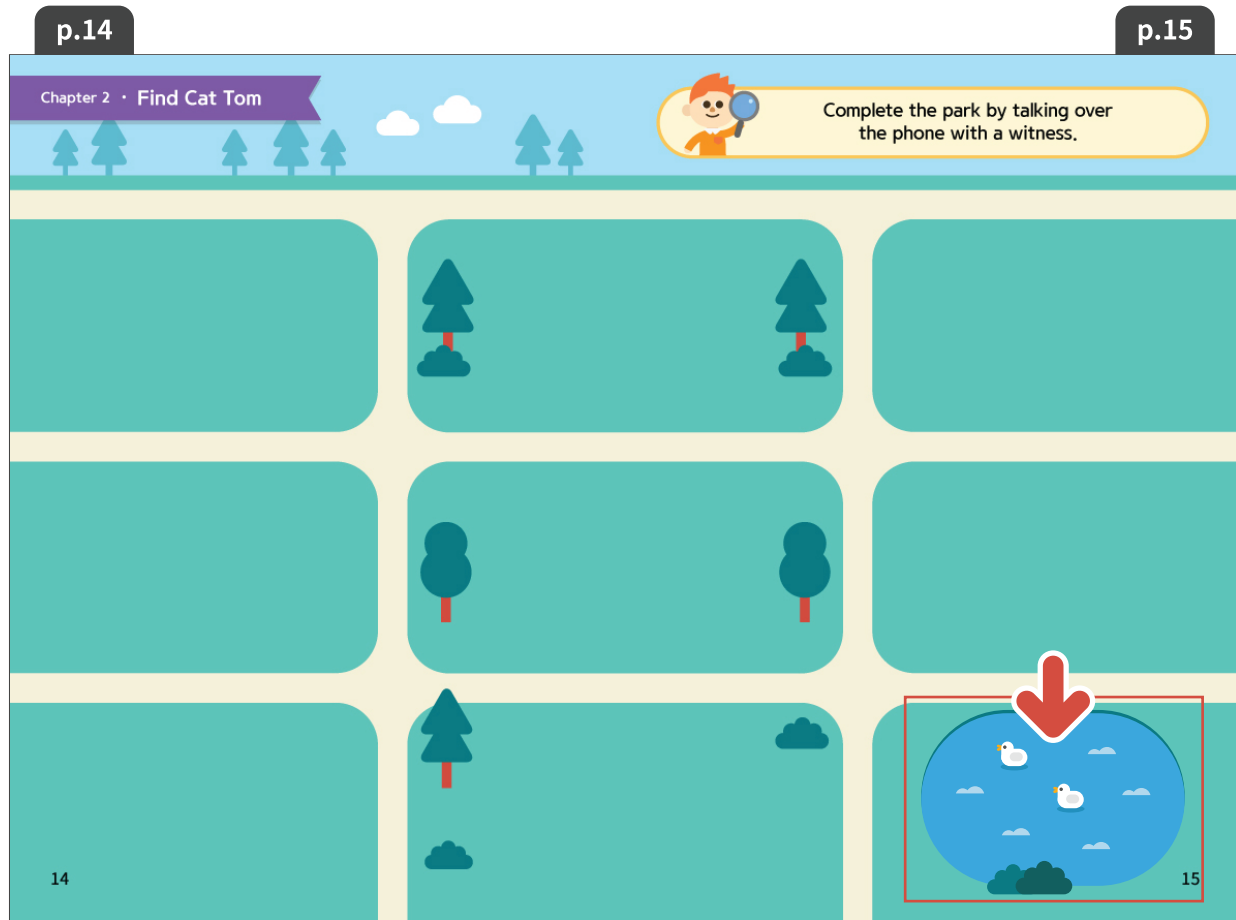


Cannot see that the cat on the first screen is wearing a pink hat or not, but you can find out that it's not the lost cat because color of tail is different.

※ Required tools : pen or pencil

Application play

Analyze characteristics of other cats and find cats hiding in the park.



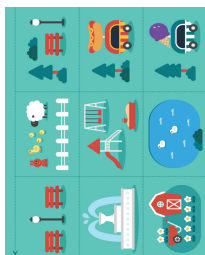
Goal

Find out structure of the park by talking on the phone with a witness, who saw the missing cat. Learn how to abstract information in this process.

How

Explain the structure of the park by setting up a situation of talking to a witness by phone. Guide to put structures in the right position on empty park based on park on page 14-15. Put park structure of supplement page on empty park. At this time, explain with words such as up, down, left, and right.

Example



“There is a pond at the bottom right of the park.”
(Encourage children to cut a pond and put in the right position.)

※ Required tools : glue, scissors

Application play

Draw more trees, grass, and etc. in empty area as you want and decorate the park.



Goal

Learn how to figure out partial characteristics of answer cat and compare with cats in the park.

How



Describe appearance in detail to identify characteristics of the missing cat.
Find out where the cat is hiding in the park by comparing hidden cats.

Application Play

Encourage to choose one thing such as animal, person, structure, or etc. in the park and to describe it. Let's others guess and get the answer.

※ Required tools : pen or pencil

p.18

Chapter 3 · Who is the criminal?

Witness 1

Yes, this is the detective office.

Someone took the cat to the bank.

Witness 2

Yes, this is the detective office.

Someone was in a cafe with the cat.

Who took the cat?
What should we ask the witnesses to find the cat?

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p.19

"Let's listen testimony of witnesses and draw clothes of a criminal."



Testimony of a witness 1

Around 3 o'clock, I saw someone who entered the bank holding a cat. That person was wearing yellow clothes.



abstract a suspect

Around 5 o'clock, I saw someone grabbing a drink with the cat. That person was wearing purple clothes, hat, necklace, and sunglasses. And that person was holding a flower.

19

How

Progress setting situation that request a case to detective.

※ Required tool : A model phone

Goal

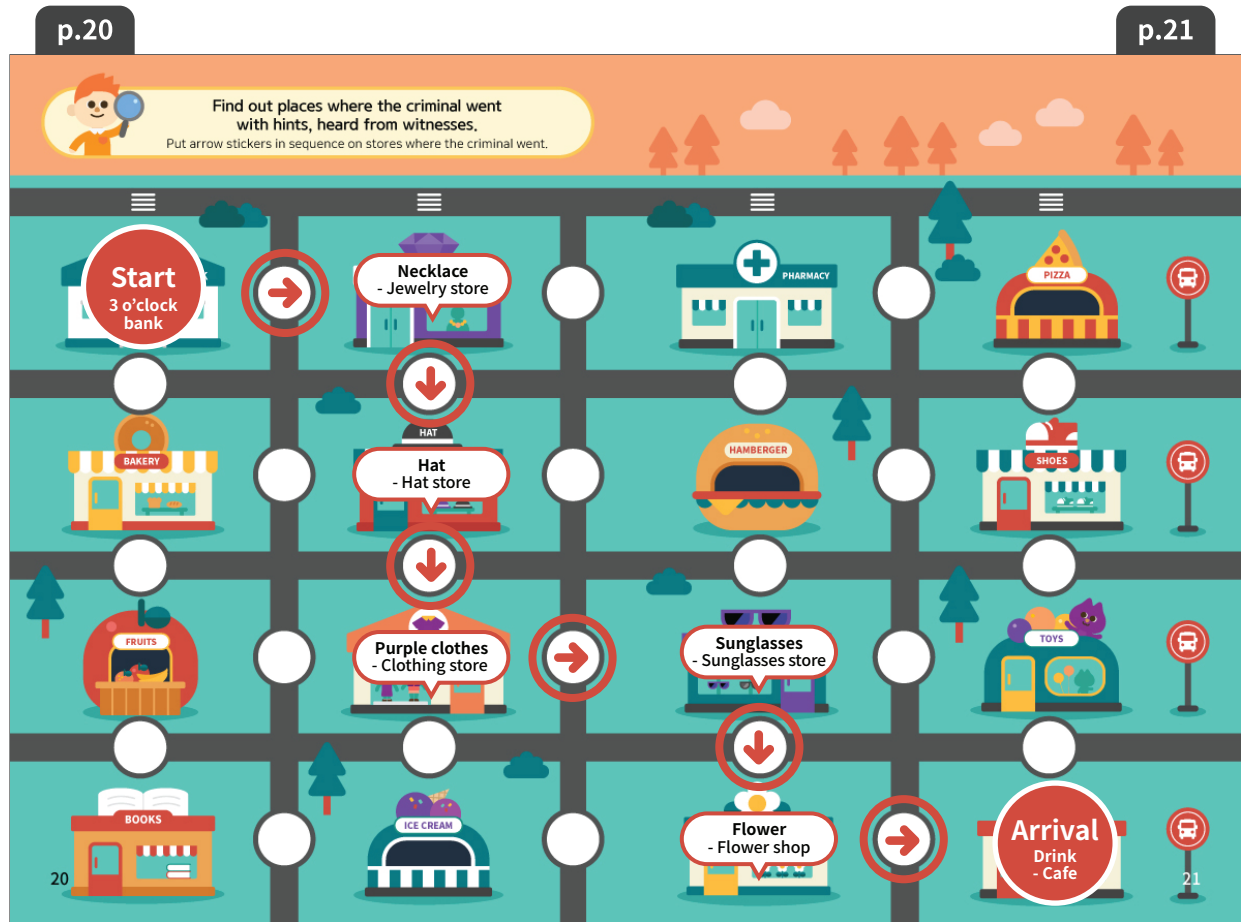
Listen testimony of a witness, figure out clothes of a criminal, and learn how to abstract information.

How

1. Paint clothes of a criminal at 3 o'clock as testimony of witness 1.
2. Paint clothes of a criminal at 5 o'clock as testimony of witness 2.
3. Encourage to find out how the criminal's clothes changed with time by comparing two testimonies.

What's different : drink, purple clothes, hat, necklace, sunglasses, flower

※ Required tools : colored pencils or crayons



Goal

Learn about '**Algorithm**', '**Sequence**', and '**Branch**' while figuring out path of a criminal by checking changed clothes.

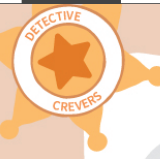
How

1. Check testimonies of a witness 1 and 2 on p.20 and find differences by observing description.
What's different : drink, purple clothes, hat, necklace, sunglasses, flower
 2. Encourage to figure out where criminal went in the village by identifying changed description.
 Branch element: yellow clothes -> purple clothes (In other words, the criminal went to a clothes store.)
 3. Let children recognize with testimony of witness on page 19 that the criminal started from the bank and encourage to identify the path.
 4. Check the path which way the criminal went by drawing arrows. At this time, explain by using words such as left, right, up, and down.
- ※ Required tools : pen or pencil

Application play

Make other paths by assuming that the criminal went other stores.

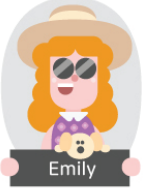
p.22



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p.23



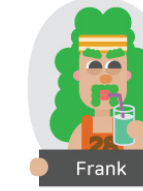
Emily



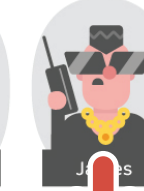
Jordan



Dr. Mueller



Frank



James



Grandmother Laura



Big bear



Ping



Daisy



Hook



Jasmine



Funk



Woody



Benjamin



Amy



Courbet



Beberu



Barbara

22

★ Detective Office ★

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Goal

Learn how to **decomposition** by observing characteristics of a criminal's description in detail.

How

Guide to find a criminal by identifying description and characteristics of a criminal who we guessed on p.19 and matching the suspects.

Characteristic : drink, purple clothes, hat, necklace, sunglasses, flower

ex) Q : Is the criminal wearing a hat? / A : Yes, the criminal is wearing a hat.

Application Play

Select one of the criminals randomly and guide to ask questions to be answered with 'yes' or 'no'.

Rule : Person who knows the criminal can only answer with 'yes' or 'no'.

ex) Q : Does the criminal have black hair? / A : No.

Guide to guess the criminal with this question.

※ Required tools : pen or pencil



Goal

Find out rule that destination is different depending on row and column in the bus ticket. (**Decomposition, Branching**)

How

Figure out secret code of bus ticket.

First row of the bus ticket means pattern and first column means color on the bus. Guide to know that destination is different depending on rows and columns.

Then, decide one destination and decorate bus on page 25.

Reference : Destination with criminal will be final destination of game on page 26-27.

Example

If destination of purple heart pattern bus is the harbor and the criminal is in the harbor, draw purple heart on the bus.

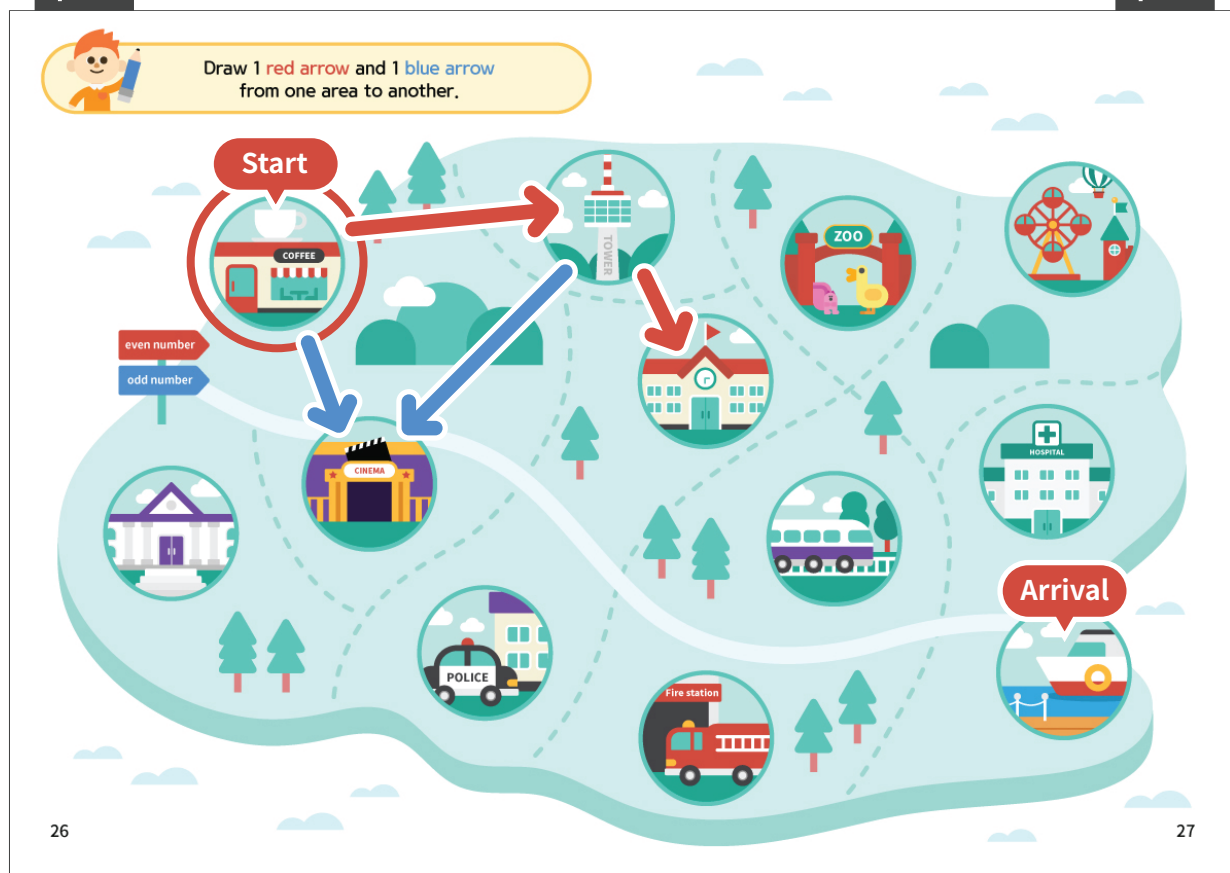
Application Play

Figure out destination of the bus with different pattern depending on rows and columns of bus ticket.

※ Required tools : pen, colored pencils or crayons

p.26

p.27



Goal

Learn about game rule by creating it yourself and progress game in sequence.
(**Sequence, Loop, Branching**)

How

Complete the board game by marking arrow so that **red line** and **blue line** can go from one area to different area.

1. Write 1~6 or any number on empty dice. (See Appendix Dice Making)
2. Starting point is cafe and arrival point is destination of the bus which we drew on p.25. (Arrival of purple heart pattern bus is the harbor.)
3. Place the piece after determining the order with rock-paper-scissors.
4. Roll the dice and when the number is even, move the horse along **red line** or when the number is odd, move the piece along **blue line**.
5. Anyone who arrives quickly to arrival point wins.

※ Required tools : red pen, blue pen, glue, scissors, dice (in Appendix), pieces (in Appendix)

Application play

Play by choosing different starting and arrival point.

create Your
universe!



create Your Universe!
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